

Grokking Under Differential Privacy

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1 Introduction

1.1 Background

Understanding why neural networks generalize as well as they do is still one of the big open questions in deep learning. A phenomenon that basically highlights this mystery is **grokking**. A model can memorize the training examples almost immediately, yet show test accuracy that is no better than random guessing. Only after a *long stretch of additional training* does it finally learn the actual structure of the task. Power et al. [1] observed this behavior on simple algorithmic tasks like modular arithmetic, where networks initially overfit but eventually reach the correct solution.

1.2 Problem Setup

In most training workflows, we would stop as soon as the validation error begins to rise, since that usually marks the transition into overfitting. **Grokking** breaks this expectation. A model can hit *near-perfect* training accuracy very early while its test accuracy stays close to random guessing for a long stretch. Only much later does it begin to show real generalization. Our work focuses on understanding this delayed transition in settings where we remove standard stabilizing tools such as weight decay and introduce new constraints through differential privacy.

1.3 Motivation

The long delay before generalization suggests that, after memorizing the training data, the model continues updating its parameters without improving its understanding of the task. Prior work has linked this behavior to issues that emerge during standard cross-entropy training [2], and has shown that weight decay can keep the model from drifting into this unproductive phase [1].

Our work examines whether the constraints introduced by differential privacy can play a similar stabilizing role. We focus on whether *clipping* and *noise*, which are built into differentially private training, can guide the model toward generalization without relying on weight decay.

1.4 Potential Implications

Whether differential privacy can stabilize grokking has direct consequences for how we think about training under constraints. If *clipping* alone prevents the numerical issues that interrupt generalization, it would show that a tool originally introduced for privacy also helps maintain stable learning.

This approach yields private models that avoid failures typically requiring *extra regularization*.

If we consider that *clipping* supports learning but the added noise limits performance, that would highlight a clear difference between restricting gradient magnitudes and adding randomness to the optimization process.

1.5 Current State of the Art

Prior work showed the delayed generalization pattern of grokking and found that weight decay plays a central role in enabling it [1]. More recent work linked the delay to numerical issues that arise during standard cross-entropy training and explained how these issues prevent the model from improving once it has memorized the data [2].

Although these studies highlighted the need for some stabilizing force during training, **they did not** examine whether the constraints introduced by differential privacy could serve that purpose.

1.6 Proposed Approach and Novelty

We study whether the mechanisms used in differentially private training can act as an alternative to weight decay for enabling grokking. Our idea is that *gradient clipping*, which limits how much each example can influence a single update, may help prevent the numerical issues that interrupt learning.

Our setup compares clipping on its own with the full private update that also includes *noise*. We start by building a numerically stable baseline from prior work. Later, we apply clipping without noise, followed by the full update at several privacy levels. This structure lets us see whether clipping supports generalization or whether the added *noise* becomes the main barrier. In addition, training data size and stochasticity are varied to see how they interact with the privacy constraints.

1.7 Our Results

In our work, we uncover a fundamental conflict between the mechanical stability provided by Differential Privacy (DP) and the delicate signal required for grokking. We demonstrate that Gradient Clipping, a core component of DP-SGD, acts as a powerful stabilizer. By restricting the update magnitude, clipping effectively substitutes for weight decay, preventing Softmax Collapse and enabling grokking even in data-scarce regimes where unregularized models fail.

However, we find that the Noise Injection required for formal privacy guarantees is fatal. Even at weak privacy budgets ($\epsilon = 20$), the noise overwhelms the subtle gradient signals required for modular addition, completely blocking the phase transition to generalization. Thus, we conclude that while the *structural* constraint of DP (clipping) is beneficial for algorithmic learning, the *informational* constraint (noise)

renders standard DP-SGD incompatible with this class of tasks.

2 Preliminaries

2.1 Grokking

A model is said to **grok** a task when it reaches near-perfect training accuracy at an early step t_{train} , while its test accuracy remains close to chance level (i.e., the accuracy obtained by uniform random guessing over all output classes). Test accuracy increases sharply at a later step t_{grok} with $t_{\text{grok}} \gg t_{\text{train}}$. Here, t_{train} marks the point where the model first memorizes the training set, and t_{grok} marks the point where generalization appears.

This pattern was first reported by Power et al. [1] on small algorithmic datasets. The main feature is the long gap between t_{train} and t_{grok} , during which the model memorizes the data but does not generalize. Test accuracy stays at chance level across this period before rising in a sharp phase transition.

2.2 Naive Loss Minimization

Naive Loss Minimization (NLM) refers to the stage of training that begins after the model reaches 100% training accuracy. At this point the gradient update tends to scale all logits in the same direction without changing the predicted label. Let $\mathbf{z} = (z_1, \dots, z_K)$ denote the vector of logits over K classes. In NLM the update often moves toward $c\mathbf{z}$ for some constant $c > 1$, which leaves the prediction unchanged since $\arg \max_k (cz_k) = \arg \max_k (z_k)$.

For the cross-entropy loss

$$\mathcal{L}_{\text{CE}} = -\log \left(\frac{e^{z_y}}{\sum_k e^{z_k}} \right)$$

where z_y is the logit of the correct class, scaling all logits lowers the loss:

$$\mathcal{L}_{\text{CE}}(c\mathbf{z}) < \mathcal{L}_{\text{CE}}(\mathbf{z}) \quad \text{for } c > 1$$

This explains why the model keeps increasing logit magnitudes even after it has memorized the data.

Prieto et al. [2] showed that in homogeneous networks this scaling effect appears naturally because gradients tend to align with the weight direction. As the weights grow, the logits grow with them, and the model is pushed toward unbounded values. This NLM behavior can drive the system toward Softmax Collapse, which blocks the transition from memorization to generalization.

2.3 Softmax Collapse

Softmax Collapse (SC) refers to the situation in which the softmax function becomes numerically unstable because the logits grow too large. When this happens, the model assigns almost all probability to a single class, and the gradients used for learning effectively vanish.

Let $\mathbf{z} = (z_1, \dots, z_K)$ be the vector of logits over K classes, and let z_y denote the logit of the correct class. SC occurs when

the softmax denominator collapses to the largest term:

$$\sum_{k=1}^K e^{z_k} \stackrel{\text{fp}}{=} e^{z_y}$$

where the symbol $\stackrel{\text{fp}}{=}$ indicates equality under finite-precision arithmetic.

Under this condition the gradient of the cross-entropy loss becomes:

$$\frac{\partial \mathcal{L}_{\text{CE}}}{\partial z_c} = \frac{e^{z_c}}{\sum_k e^{z_k}} - 1_{c=y} \stackrel{\text{fp}}{=} 1 - 1_{c=y}$$

For the correct class ($c = y$), this gradient becomes zero, so the model stops receiving useful learning signal from that example.

In practice SC appears when the logits become extremely large. Hence, the softmax output saturates, the gradients shrink, causing the model to no longer refine its learned representations.

2.4 StableMax

StableMax is a modification of the softmax function designed to avoid the floating-point issues that lead to Softmax Collapse. It replaces the exponential terms with a piecewise linear function that stays well behaved even when the logits grow large.

The underlying function is defined as:

$$s(x) = \begin{cases} x + 1, & x \geq 0 \\ \frac{1}{1-x}, & x < 0 \end{cases}$$

Using $s(\cdot)$, the StableMax output for the logit z_i is:

$$\text{StableMax}(z_i) = \frac{s(z_i)}{\sum_j s(z_j)}$$

The corresponding StableMax cross-entropy loss is:

$$\mathcal{L}_{\text{StCE}}(f(x), y) = -\log(\text{StableMax}(z_y))$$

where z_y is the logit of the correct class.

The idea behind StableMax is that $s(x)$ grows linearly for nonnegative inputs and does not shrink too quickly for negative inputs. This avoids the extreme ratios produced by the exponential function, preventing the absorption errors that occur when one logit dominates the others. Prieto et al. [2] explained that this stability is often enough to enable grokking without weight decay, making StableMax a useful baseline in our experiments.

2.5 Differential Privacy

Differential privacy describes how much the output of an algorithm can change when a single example in the dataset is modified. Replacing one data point should not noticeably affect the distribution of outputs produced by the algorithm.

Let $\mathcal{M}$ be a randomized algorithm. It is said to satisfy (ϵ, δ) -differential privacy if the following condition holds for

any two datasets D and D' that differ in exactly one entry, and for any measurable set S :

$$\Pr[\mathcal{M}(D) \in S] \leq e^\epsilon \Pr[\mathcal{M}(D') \in S] + \delta$$

The parameter ϵ controls how sensitive the output can be to a single example, with smaller values giving stronger privacy. The parameter δ allows for a small chance that this guarantee may not hold.

2.6 DP-SGD Algorithm

The DP-SGD update follows the same basic structure as standard SGD but adds two modifications. The first limits how large each individual gradient can be, and the second adds noise to the average of the clipped gradients. These changes ensure that no single example can strongly influence the update, which is what allows the method to satisfy a chosen privacy level.

The first step is *per-example gradient clipping*. For each example i with gradient g_i , the clipped gradient is defined as:

$$\tilde{g}_i = \frac{g_i}{\max\left(1, \frac{\|g_i\|_2}{C}\right)}$$

where C is a fixed clipping threshold. This ensures that $\|\tilde{g}_i\|_2 \leq C$ for all examples.

The second step *averages the clipped gradients* and adds Gaussian noise:

$$g = \frac{1}{n} \sum_{i=1}^n \tilde{g}_i + \mathcal{N}(0, \sigma^2 C^2 I),$$

where n is the batch size and σ is the noise scale. The value of σ is chosen so that the overall procedure meets a specified (ϵ, δ) privacy guarantee. The clipping threshold C limits the sensitivity of the update before noise is added.

2.7 Results from Prior Work

Several findings from earlier work guide our setup and help explain the behavior we observe in our experiments.

Result 1: Softmax Collapse blocks generalization

Prieto et al. [2] showed that without any form of regularization, Naive Loss Minimization pushes the logits to grow without bound. This eventually leads to Softmax Collapse, at which point the gradients for correctly classified examples vanish and the model cannot move from memorization to generalization.

Result 2: Weight decay prevents collapse

Weight decay keeps the logits from diverging by discouraging large parameter norms. This avoids the numerical issues that lead to collapse and allows the model to continue receiving a useful learning signal [2].

Result 3: Gradient clipping limits sensitivity

In DP-SGD, clipping ensures that each per-example gradient satisfies $\|\tilde{g}_i\|_2 \leq C$. This follows directly from the clipping rule introduced in Section 2.6 and prevents any single example from producing an unusually large update.

3 Related Work

3.1 Power et al. (2022)

Power et al. [1] introduced *grokking* through small algorithmic tasks and showed that weight decay often plays an important role in allowing delayed generalization to appear. Their setup provides the baseline setting that our work builds on.

3.2 Prieto et al. (2025)

Prieto et al. [2] linked the absence of *grokking without weight decay* to numerical issues, particularly Softmax Collapse. They also proposed StableMax as a stable alternative that avoids these failures. Their results motivate our interest in whether other constraints, such as differential privacy, can have a similar stabilizing effect.

3.3 Additional Related Work

3.3.1 Varma et al. (2023)

Varma et al. [3] argued that *grokking* reflects a shift from early memorization to a more efficient circuit for generalization. They suggest that this shift is delayed because the more efficient solution is harder for the model to reach. Their viewpoint helps explain the long plateau before test accuracy improves.

3.3.2 Lee et al. (2024)

Lee et al. [4] studied the *relative size of gradients* responsible for memorization versus generalization. They found that generalizing gradients are often much smaller, which makes them easy to overlook during training. This observation was useful for our work because DP noise may suppress these smaller gradient components.

3.3.3 Koloskova et al. (2023)

Koloskova et al. [5] experimented how *gradient clipping* changes the behavior of stochastic gradient descent. They showed that clipping introduces bias into the gradient estimate, which can slow convergence. Since clipping is a central part of DP-SGD, their results raise the question of whether clipping alone might delay the onset of *grokking*.

3.3.4 Cohen et al. (2021)

Cohen et al. [6] showed that gradient descent often operates near the *edge of stability*. Their results give context for whether DP-SGD changes the training dynamics enough to influence when *grokking* emerges.

4 Methodology

To investigate the relationship between differential privacy and *grokking*, we adopted the experimental framework established by [1] and extended it with the stability analysis tools from [2]. This section details our dataset configuration, model architecture, and the specific DP-SGD interventions.

4.1 Task and Dataset Configuration

We focused on the algorithmic task of **Modular Addition**, defined as learning the operation $a + b \pmod{p}$ from a subset of example pairs.

- **Problem Space:** We set the modulus $p = 113$ (a prime number), resulting in a total dataset of $113^2 = 12,769$

unique equations.

- **Input Representation:** Integers a and b are represented as distinct tokens and embedded into high-dimensional vectors. The model treats the task as a sequence-to-sequence problem (predicting c given $a, b, =$).
- **Data Regimes:** To capture the grokking behavior, we utilized data splitting fractions that sit on the boundary of learnability:
 - *Data-Scarce* (40%) : A difficult situation where standard models typically suffer from Softmax Collapse and fail to generalize.
 - *Data-Rich* (60%) : A comparatively easier situation where generalization is faster, allowing us to test if DP methods actively hinder learning.

4.2 Model Architecture

We employed a standard decoder-only **Transformer** architecture, consistent with prior literature on algorithmic grokking [2].

- **Specs:** 2 layers, 4 attention heads, and a hidden dimension of 128.
- **Precision:** All baseline experiments were conducted using `float64` precision to avoid numerical errors, so that Softmax Collapse appears as a genuine effect and not a consequence of limited precision.

4.3 Experiment Configurations

Our experiments were designed to separate the two parts of Differential Privacy: *gradient clipping* (a structural constraint) and *noise injection* (an informational constraint).

4.3.1 Numerical Stability Baselines

To benchmark our results, we established two non-private baselines:

- **Standard Cross-Entropy (CE):** Trained with *AdamW*. This serves as the baseline for observing the onset of Softmax Collapse.
- **StableMax:** Trained using the numerically stable *StableMax* loss function proposed by [2]. This serves as the upper bound for performance, showing ideal grokking behavior under perfect numerical stability.

4.3.2 The Clipping-Only Ablation

To test the hypothesis that clipping acts as a regularizer, we implemented a custom optimizer that applies per-example Gradient Clipping (max norm $C = 1.0$) **without** adding noise. This isolates the geometric effect of bounding the gradient updates from the stochastic effect of privacy noise.

4.3.3 Differentially Private SGD (DP-SGD)

We implemented full DP-SGD using the *Opacus* library. The clipping norm was fixed at $C = 1.0$, and the noise multiplier σ was varied to achieve two distinct privacy budgets:

- **Moderate Privacy** ($\epsilon = 8$): Testing feasibility under standard privacy guarantees.
- **Weak Privacy** ($\epsilon = 20$): A high-budget setting to test if the model can learn even under minimal noise conditions.

Note: Privacy accounting was performed using the RDP (*Rényi Differential Privacy*) accountant with $\delta = 10^{-5}$.

4.4 Evaluation Metrics

We trained all configurations for up to 100,000 epochs. We monitored four key metrics to diagnose training behavior:

1. **Test Accuracy:** Grokking is defined as the phase transition where test accuracy rises from random chance ($\approx 1/p$) to above 95%.
2. **L2 Weight Norm:** To track the onset of Naïve Loss Minimization (NLM), where weights grow uncontrollably to squash the loss.
3. **Cosine Similarity:** We measure the cosine similarity between the model’s weight vector and the gradient update. A strong negative alignment (≈ -1.0) indicates the model is minimizing loss purely by scaling up logits (NLM) rather than learning features [2].
4. **Zero Gradient Percentage:** The proportion of parameter updates in a batch that are numerically zero. This serves as a direct proxy for Softmax Collapse, indicating that floating-point saturation has destroyed the gradient signal.

5 Evaluation Results

In this section, we present the empirical results of our investigation. We first establish the validity of our experimental pipeline by replicating recent baseline findings regarding numerical stability in grokking. Then, we isolate the specific mechanisms of Differentially Private SGD (DP-SGD) i.e gradient clipping and noise injection, to evaluate their individual and combined impacts on the grokking behavior.

Unless otherwise stated, all experiments were conducted using a 2-layer MLP trained on the Modular Addition dataset ($P = 113$) with a 40% data fraction and the *AdamW* optimizer.

5.1 Baseline Replication: Numerical Stability vs. Softmax Collapse

To establish a reliable baseline, we first replicated the core finding of [2]. Their work posits that the delayed generalization characteristic of grokking is often hindered not by a lack of regularization, but by numerical instabilities arising from **Naïve Loss Minimization (NLM)**. For example, as models overfit, logit magnitudes grow uncontrollably, causing floating-point errors in the standard Softmax function, a phenomenon termed **Softmax Collapse**.

We compared the training dynamics of two identical MLPs trained **without weight decay** ($\lambda = 0$). One model used the standard Cross-Entropy loss with Softmax, while the other used the proposed **StableMax** activation, designed to handle large logits without numerical failure. The results, shown in Figure 1, confirm the baseline findings.

The model trained with standard Softmax (Blue curves) successfully memorizes the training data, achieving 100% training accuracy very early. However, it completely fails to generalize, and the test accuracy remains stuck at random chance.

Importantly, the baseline’s **Softmax Collapse** metric (Cyan dotted line) spikes significantly and remains high. This confirms that the model has entered a numerically unstable state where gradients are vanishing due to floating-point saturation. Due to this observed instability and the lack of learning progress, we terminated the baseline run at 20,000 epochs, as further training in this collapsed regime is unproductive.

In contrast, the model trained with StableMax (Red curves) maintains numerical stability (zero collapse). While it also quickly memorizes the training data, it eventually undergoes a distinct **phase transition** around epoch 75,000, where test accuracy rapidly rises to 100%. This confirms that explicit regularization is not strictly necessary for generalization, provided numerical stability is maintained.

5.2 The Stabilizing Effect of Gradient Clipping

Having established that numerical instability hinders grokking, we explored whether **Gradient Clipping**, a core component of DP-SGD, could independently stabilize training. Our hypothesis was that clipping acts as a hard constraint on the update magnitude during Naïve Loss Minimization (NLM), restricting the growth of weight norms and preventing the logits from reaching the magnitude required to trigger Softmax Collapse.

We trained the model using standard Cross-Entropy loss but applied gradient clipping (maximum norm $C = 1.0$) without adding any privacy noise ($\sigma = 0$). We evaluated this change across two distinct data cases.

Scenario 1: Data-Rich (60% Training Fraction)

In this easier regime, the abundance of data provides a stronger gradient signal. We tested two batch sizes to separate the effects of SGD-induced stochasticity from the clipping mechanism. Figures 2 and 3 demonstrate that in a data-rich environment, gradient clipping is benign.

Whether using small batches (high noise) or large batches (low noise), the clipping constraint does not prevent the model from converging to a generalized solution.

Scenario 2: Data-Scarce (40% Training Fraction)

This regime represents the edge of stability where grokking typically fails. As shown in Figure 4, the difference is clear.

In this setting, the unregularized baseline (Blue) suffers from Softmax Collapse; it memorizes the training data but the test accuracy flatlines at random chance, and the L2 norm of the weights increases rapidly (top-right subplot).

In contrast, the model with gradient clipping (Red) successfully undergoes a phase transition to perfect test accuracy. The L2 norm plot confirms the mechanism: while the baseline weights grow unchecked, the clipped weights are constrained. This constraint prevents the logits from triggering numerical instability, validating our hypothesis that **Gradient Clipping alone is sufficient to induce grokking** by

mechanically enforcing stability.

5.3 The Destabilizing Effect of Privacy Noise

While Gradient Clipping provides the necessary numerical stability (as demonstrated in Section 5.2), the full DP-SGD algorithm also requires **Noise Injection** to satisfy the differential privacy guarantee. We checked whether the model could still grok under the influence of this noise.

We utilized the data-rich regime (60% training fraction, batch size 256) where the baseline model generalized easily. We swept over privacy budgets $\epsilon \in \{8, 20\}$. Note that $\epsilon = 20$ represents a very high privacy budget (low noise), which provides the easiest possible condition for the model to learn under DP.

Moderate Privacy ($\epsilon = 8$)

First, we tested a moderate privacy budget of $\epsilon = 8$. As shown in Figure 5, the results indicate a complete stopping of learning.

The DP Softmax Collapse metric (purple dotted line) remains at zero throughout the run, confirming that our hypothesis about clipping was correct: it successfully prevented the numerical instability that plagues the baseline. However, the Red curves (DP Train and Test Accuracy) remain flat at random chance. The noise magnitude required for $\epsilon = 8$ appears to completely drown out the gradient signal for this modular arithmetic task.

Weak Privacy ($\epsilon = 20$)

To determine if this failure was simply due to excessive noise, we relaxed the constraint significantly to $\epsilon = 20$. As seen in Figure 6, the outcome remains unchanged.

Even with the minimal noise associated with $\epsilon = 20$, the model fails to memorize or generalize.

6 Summary and Future Work

Our study of Grokking under Differential Privacy shows a clear difference between the *structural* benefits of Gradient Clipping and the *information loss* introduced by Privacy Noise.

6.1 Summary of Findings

Our experiments reveal a critical **hierarchy of constraints** in grokking:

- **Primary Bottleneck (Stability):** For unregularized models, the immediate barrier to generalization is numerical instability (Softmax Collapse). We proved that **Gradient Clipping** successfully cures this by mechanically restricting weight updates, substituting for weight decay.
- **Secondary Bottleneck (SNR):** Once stability is achieved, the **Signal-to-Noise Ratio (SNR)** becomes the limiting factor. Under DP-SGD, the noise level overwhelmed the gradient signal.

Despite the stability provided by clipping, the injection of Gaussian noise, even at weak privacy levels ($\epsilon = 20$), prevented the model from generalizing. This suggests that the gradient features required to learn modular arithmetic are very small (low magnitude) and are destroyed by the noise levels required

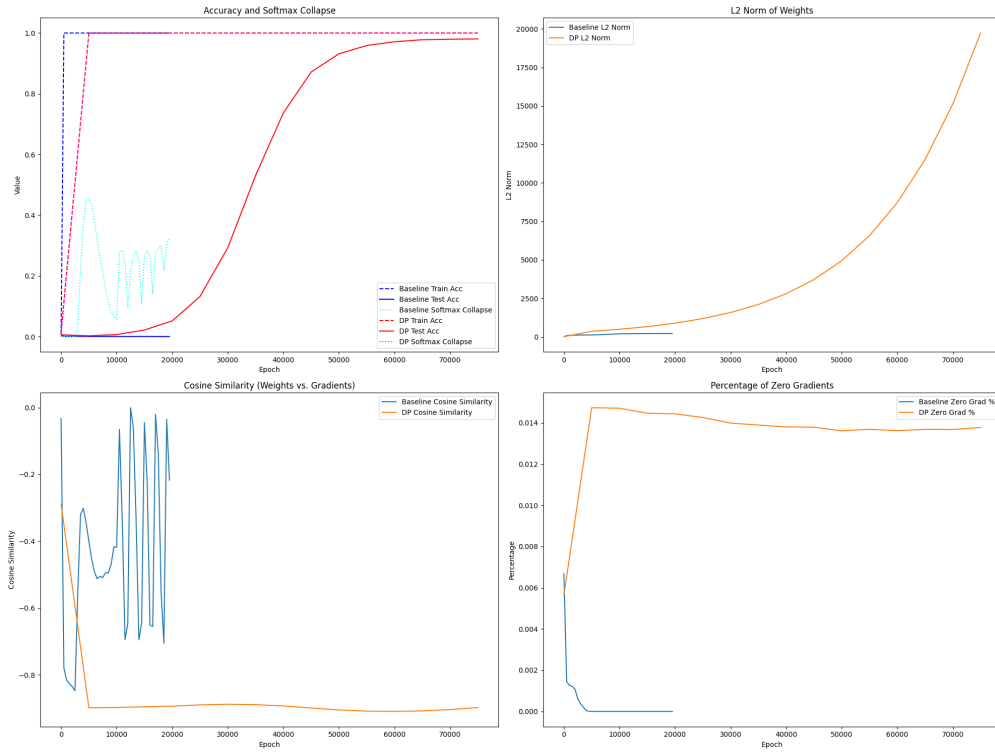


Figure 1: **Baseline Replication (No Weight Decay).** The Baseline model (Blue) successfully memorizes the training data (100% Train Acc) but fails to generalize (Test Acc stays near 0). The Softmax Collapse metric (Cyan dotted) rises sharply, indicating numerical failure. Whereas, the StableMax model (Red) maintains stability, eventually **grokking** around epoch 80,000. *Note: Due to a plotting artifact, the legend labels StableMax as DP.*

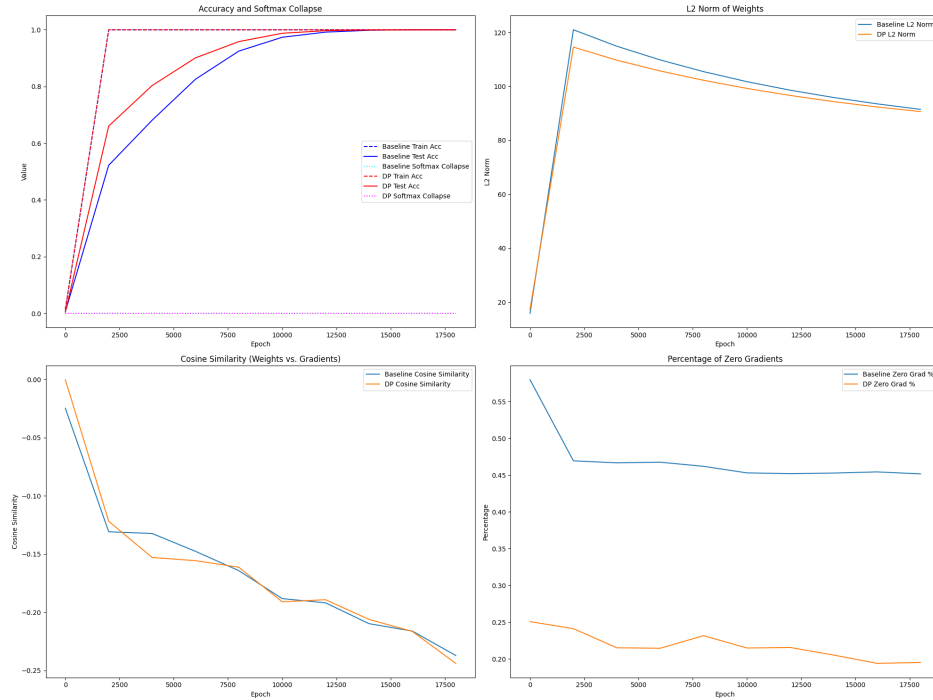


Figure 2: **Data-Rich Regime (60% Data, Batch 256).** With high stochasticity (small batch size), both the Baseline (Blue) and Clipping-Only model (Red) successfully grok. This confirms that in favorable conditions, gradient clipping does not impede the learning of generalized features.

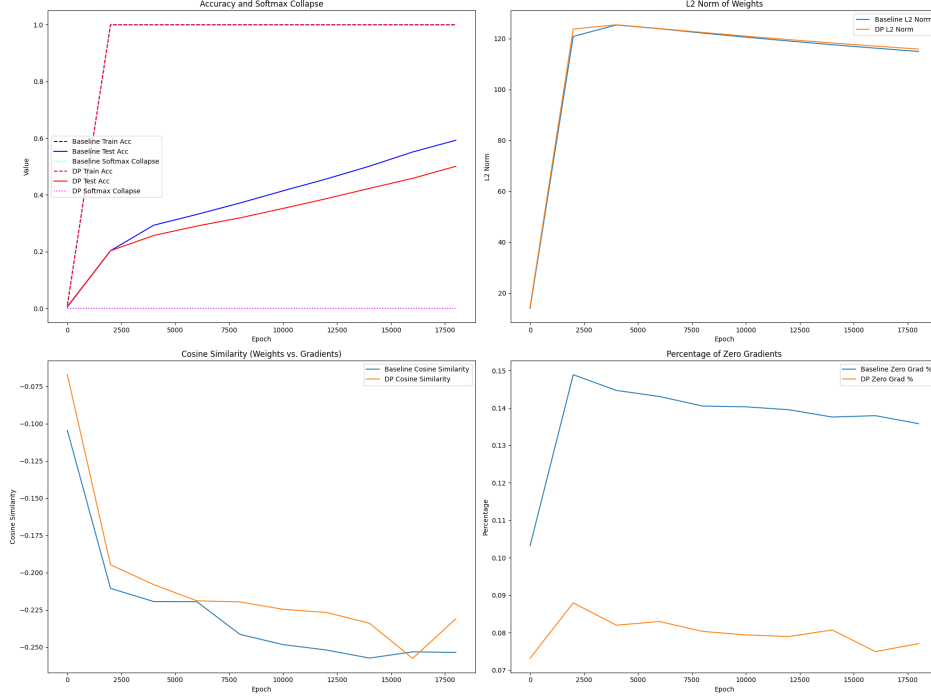


Figure 3: **Data-Rich Regime (60% Data, Batch 1024)**. With low stochasticity (large batch size), the Baseline (Blue) struggles more compared to the smaller batch setting, showing delayed progress. The Clipping-Only model (Red) maintains a competitive learning trend, suggesting that clipping may partly compensate for reduced SGD noise even in high-data regimes.

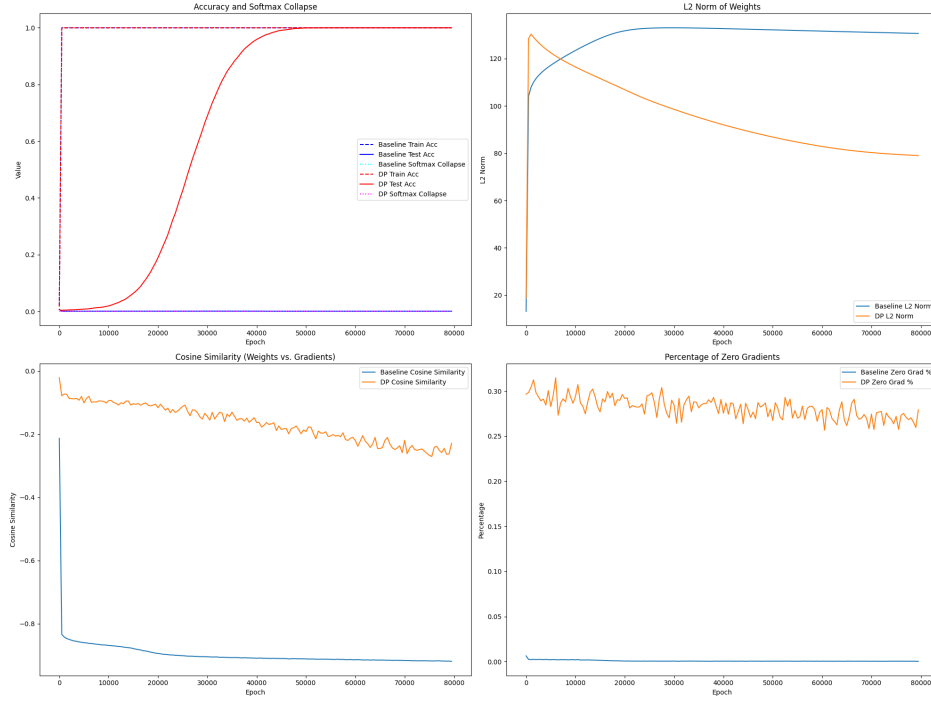


Figure 4: **Data-Scarce Regime (40% Data, Batch 1024)**. This is the important observation. The Baseline (Blue) memorizes the data (dashed line) but **never generalizes** (solid line stays at 0), as NLM dominates and the L2 weight norm increases rapidly (top right). The Clipping-Only model (Red) **successfully groks** around epoch 40,000. The weight norm plot confirms that clipping constrains the parameter growth, preventing Softmax Collapse and preserving the weak generalization signal.

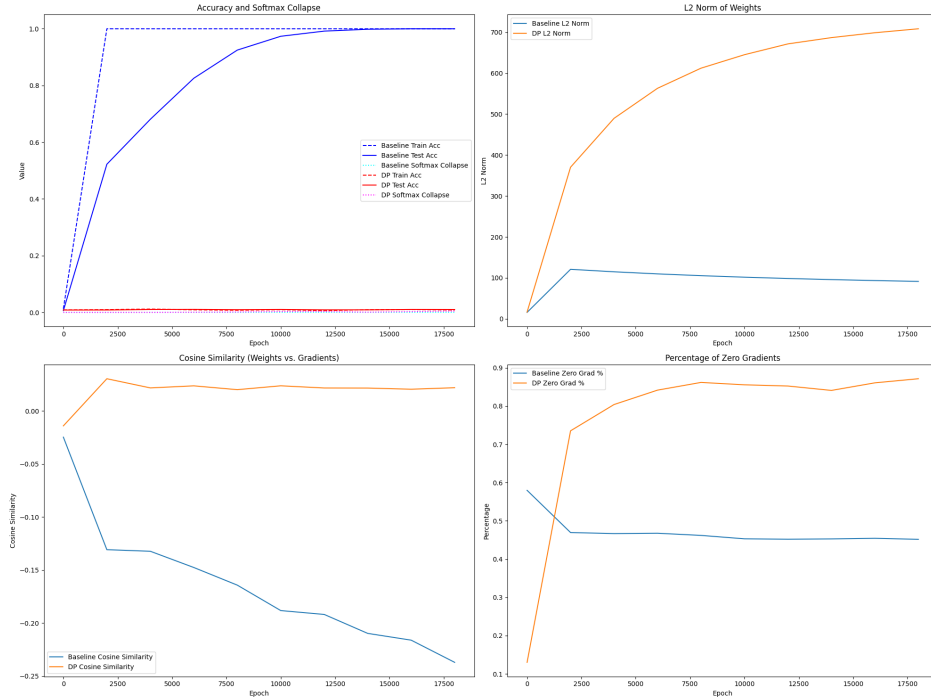


Figure 5: **DP Failure at $\epsilon = 8$ (60% Data).** The Baseline (Blue) learns quickly. The DP model (Red), however, flatlines. The DP Softmax Collapse (purple dotted line) remains at zero, confirming that **clipping successfully prevented instability**. However, the noise prevented even the initial phase of memorization (DP Train Acc is near zero).

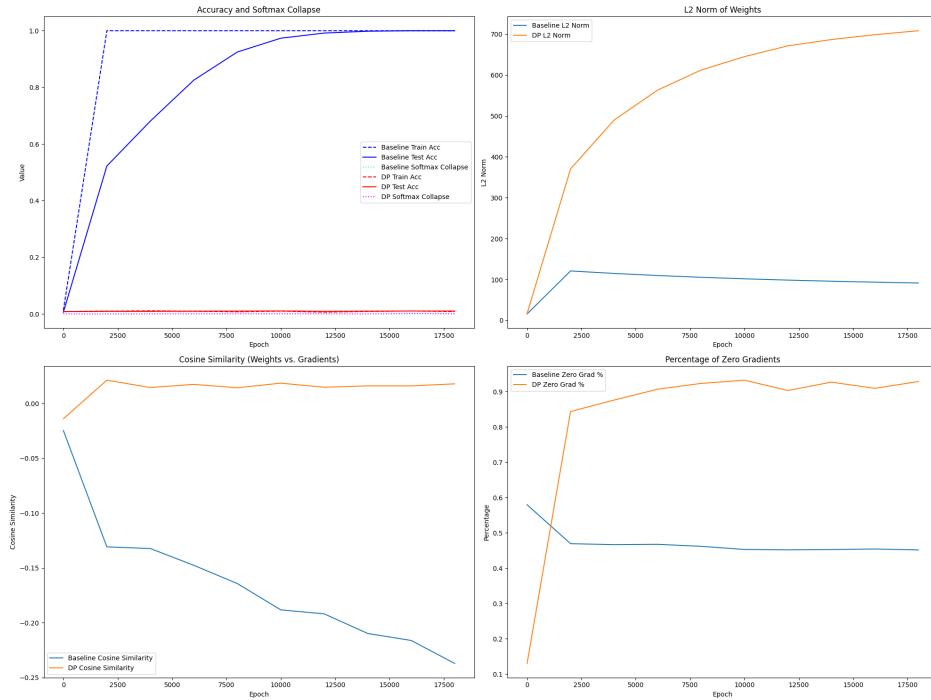


Figure 6: **DP Failure at $\epsilon = 20$ (60% Data).** Even with a very permissive privacy budget (low noise), the model fails to learn. While the clipping mechanism maintains numerical stability (purple line at 0), the gradient signal for modular arithmetic appears too fragile to survive even minimal noise injection.

for even weak differential privacy.

6.2 Future Directions

To further separate the relationship between privacy and grokking, we propose two targeted directions for future research:

- **Warm-Start Collapse Test:** Initialize a model that has already grokked (via Clipping-Only) and continue training with DP-SGD. This would determine if privacy noise merely prevents the *discovery* of features (optimization difficulty) or actively destroys *existing* ones.
- **SNR Analysis:** Systematically vary the training data fraction and batch size to map the signal strength of the generalizing gradients. The goal is to identify an Optimal Zone where the gradient signal is strong enough to withstand the DP noise floor.

7 Conclusion

This work demonstrates that while **Gradient Clipping** offers a powerful tool for numerical stability, capable of inducing grokking without weight decay, the associated **Privacy Noise** presents a strong obstacle. The transition from memorization to generalization appears to require a gradient fidelity that is currently incompatible with standard differential privacy guarantees on small algorithmic tasks. Future work must focus on amplification of the generalization signal to bridge this gap.

References

- [1] Power, A., Burda, Y., Edwards, H., Babuschkin, I., & Misra, V. (2022). *Grokking: Generalization Beyond Overfitting on Small Algorithmic Datasets*. arXiv:2201.02177.
- [2] Prieto, L., Barsbey, M., Mediano, P. A. M., & Birdal, T. (2025). *Grokking at the Edge of Numerical Stability*. ICLR 2025.
- [3] Varma, V., Shah, R., Kenton, Z., Kramár, J., & Kumar, R. (2023). *Explaining Grokking Through Circuit Efficiency*. arXiv:2309.02390.
- [4] Lee, J., Kang, B., & Kim, K. (2024). *Grokfast: Accelerated Grokking by Amplifying Slow Gradients*. ICML 2024.
- [5] Koloskova, A., Stich, S. U., & Jaggi, M. (2023). *Revisiting Gradient Clipping: Stochastic Bias and Tight Convergence Guarantees*. NeurIPS 2023.
- [6] Cohen, J., Kaur, S., Li, Y., Kolter, J. Z., & Talwalkar, A. (2021). *Gradient Descent on Neural Networks Typically Occurs at the Edge of Stability*. ICLR 2021.